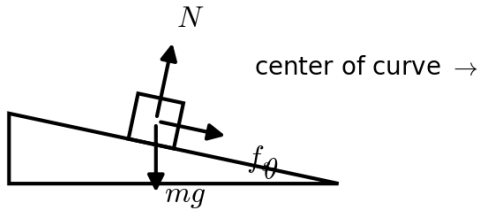
# PHYS1408 Practice Exam 2-4

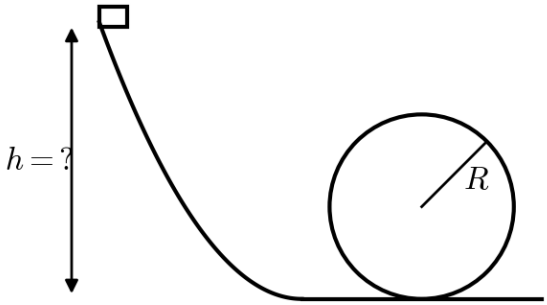
**Chapters 6, 7/8, 9, 10, 11**

Use $g = 9.8$ m/s$^2$. Choose the single best answer. Answers are rounded to the precision shown. For a minimum or maximum, choose the answer choice that actually works.

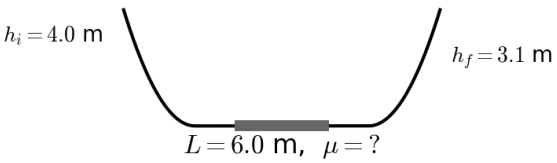
**1.** A highway curve of radius 45 m is banked at $\theta = 12°$. The coefficient of static friction between the tires and the road is $\mu_s = 0.30$. What is the maximum speed at which a car can take the curve without sliding up the bank?
a) 9.7 m/s b) 11.5 m/s c) 15.5 m/s d) 21.9 m/s
e) 241 m/s



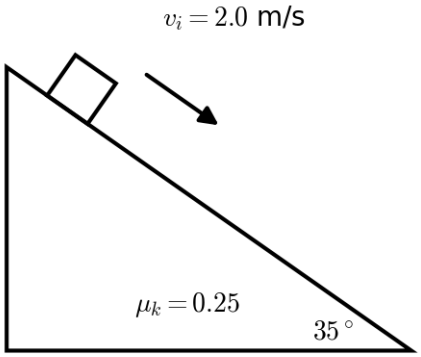

**2.** A roller coaster car starts from rest on a frictionless track and enters a vertical circular loop of radius $R = 7.5$ m. What is the minimum starting height (measured from the bottom of the loop) for the car to make it all the way around the loop?
a) 11 m b) 15 m c) 18 m d) 20 m e) 38 m



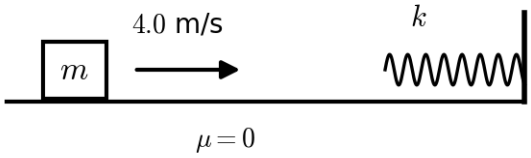

**3.** A skateboarder starts from rest 4.0 m up one wall of a halfpipe. The halfpipe is frictionless except for a 6.0 m long rough section of the flat bottom. After one pass across the rough section, she rises to a height of 3.1 m on the far wall. What is the coefficient of kinetic friction of the rough section?
a) 0.15 b) 0.52 c) 0.67 d) 0.78 e) 0.90



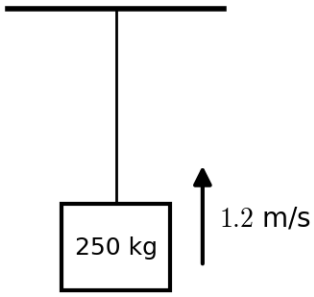
**4.** A block slides **down** a 35° incline ($\mu_k = 0.25$) with an initial speed of 2.0 m/s. How fast is it moving after it has slid 3.0 m along the incline?
a) 2.0 m/s b) 3.85 m/s c) 4.66 m/s d) 4.80 m/s
e) 5.07 m/s



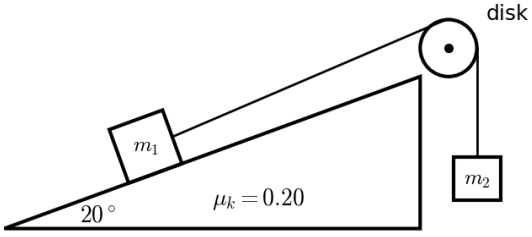

**5.** A 1.5 kg block slides at 4.0 m/s along a frictionless floor into a spring with $k = 600$ N/m. How far is the spring compressed when the block comes to rest?
a) 0.010 m b) 0.040 m c) 0.14 m d) 0.20 m e) 0.28 m



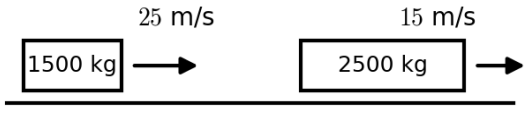

**6.** A motor lifts a 250 kg crate straight up at a constant speed of 1.2 m/s. What power does the motor deliver?
a) 180 W b) 300 W c) 2940 W d) 5880 W
e) 29,400 W



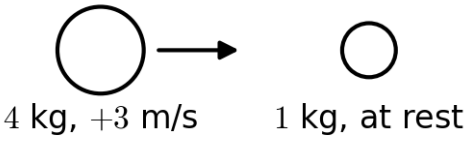

**7.** A 6.0 kg block on a 20° incline ($\mu_k = 0.20$) is tied by a light string over a pulley to a 5.0 kg hanging mass. The pulley is a uniform disk ($I = \frac{1}{2}MR^2$, $M = 4.0$ kg). The system is released from rest, and the hanging mass falls. How fast is it moving after it has fallen 1.2 m?
a) 1.69 m/s b) 1.81 m/s c) 1.97 m/s d) 2.31 m/s
e) 4.85 m/s



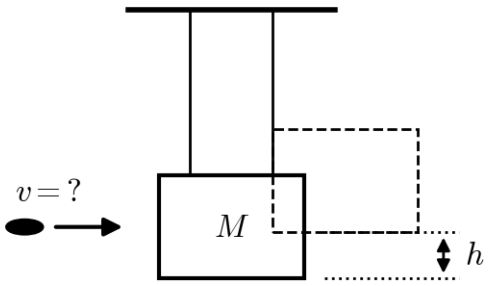

**8.** A 1500 kg car moving at 25 m/s rear-ends a 2500 kg truck moving at 15 m/s in the same direction. They lock together. What is their speed just after the collision?
a) 0 m/s b) 9.4 m/s c) 18.8 m/s d) 20.0 m/s
e) 40.0 m/s



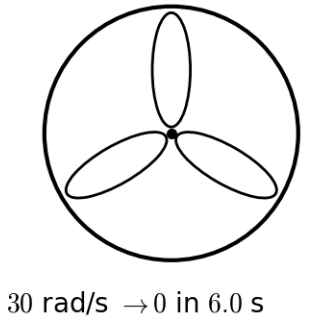

**9.** A 4.0 kg ball moving to the right at 3.0 m/s collides **elastically** head-on with a 1.0 kg ball at rest. What is the velocity of the 1.0 kg ball after the collision?
a) 0 m/s b) +1.8 m/s c) +2.4 m/s d) +3.0 m/s
e) +4.8 m/s



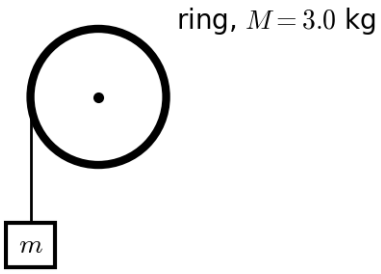
**10.** A 0.012 kg bullet is fired into a 1.5 kg block of wood hanging at rest from light strings. The bullet lodges in the block, which swings up to a height of 0.12 m. What was the bullet's speed?
a) 1.5 m/s b) 17 m/s c) 126 m/s d) 193 m/s
e) 386 m/s



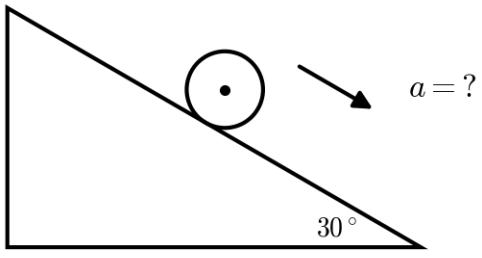

**11.** A ceiling fan spinning at 30 rad/s is switched off and slows uniformly to a stop in 6.0 s. Through what angle does it turn while stopping?
a) 5.0 rad b) 14.3 rad c) 15 rad d) 45 rad e) 90 rad



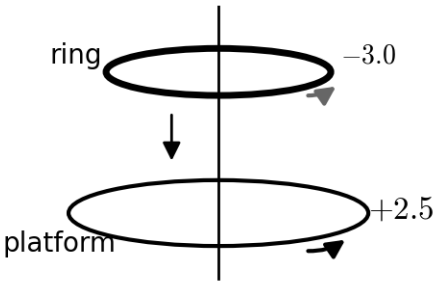

**12.** A 2.0 kg mass hangs from a light string wrapped around a pulley that is a thin ring ($I = MR^2$, $M = 3.0$ kg) on a frictionless axle. The mass is released. What is its acceleration?
a) 2.45 m/s$^2$ b) 3.92 m/s$^2$ c) 5.60 m/s$^2$ d) 6.53 m/s$^2$
e) 9.80 m/s$^2$




**13.** A solid cylinder ($I = \frac{1}{2}MR^2$) rolls without slipping down a 30° incline. What is the acceleration of its center?
a) 2.45 m/s$^2$ b) 3.27 m/s$^2$ c) 3.50 m/s$^2$ d) 4.90 m/s$^2$
e) 6.53 m/s$^2$




**14.** A platform (a disk, $I = \frac{1}{2}MR^2$, $M = 150$ kg, $R = 2.0$ m) rotates at +2.5 rad/s. A thin ring ($I = mr^2$, $m = 40$ kg, $r = 1.5$ m) spinning in the **opposite** direction at −3.0 rad/s is dropped centered onto it, and the two end up turning together. What is their final angular velocity?
a) +1.23 rad/s b) +1.78 rad/s c) +1.92 rad/s
d) +2.50 rad/s e) +2.62 rad/s



**15.** A 60 kg person stands at the **center** of a merry-go-round (a disk, $I = \frac{1}{2}MR^2$, $M = 250$ kg, $R = 2.0$ m) that rotates at 1.8 rad/s. The person walks out to the rim. What is the angular velocity when the person reaches the rim? (Treat the person as a point mass.)

a) 1.22 rad/s  b) 1.45 rad/s  c) 1.80 rad/s  d) 2.23 rad/s
e) 2.66 rad/s

